IndiaWorks Workforce Taxonomy Database & GraphQL Architecture

India-focused canonical design for blue, white, pink, grey/black-collar labels, gig/platform work, occupations, skills and enterprise candidate search

Research refresh: 25 September 2026 | Version 2.0

1. Architecture decision

Decision: Do not use collar colour or gig status as the root of the occupation tree. Use a multi-axis model anchored on India's formal occupation and skill frameworks, with collar labels retained as secondary market tags.

- NCO-2015 is the canonical occupation reference. It uses an 8-digit hierarchy aligned to ISCO-08 and covers 3,600 civilian occupations across 52 sectors [S1][S2].
- NIC-2025 is the current industry/economic-activity reference. Employer industry belongs on Organisation/JobPosting, not inside occupation identity [S3].
- NSQF-2023, NOS and Qualification Packs define competency and qualification structure. They complement occupation rather than replace it [S4][S5][S6].
- Gig/platform work is an engagement relationship. The Code on Social Security defines gig work outside the traditional employer-employee relationship; the same occupation may be permanent, contract, gig, self-employed or platform-mediated [S8].
- e-Shram explicitly distinguishes occupation from skill and supports primary plus secondary occupations, which supports a many-to-many Candidate-to-Occupation model [S9][S10].
- Blue/white/pink/grey/black-collar terminology is descriptive and overlapping. It is not the Indian statutory occupation hierarchy. Pink is historically gender-associated; black-collar is particularly ambiguous. Store these only as optional tags/analytics classifications [S13][S14][S15].

2. Canonical classification stack

```
Organisation Industry (NIC-2025)
    |
    ├── Job Family (product taxonomy)
    |     └── Canonical Role / Occupation
    |           └── Specialisation
    |                 └── Work Context / Service Context
    |
    ├── External Occupation Mapping (NCO-2015 / ISCO-08)
    ├── Competency Mapping (NQR / NSQF / QP / NOS)
    ├── Skills / Tools / Equipment / Languages
    ├── Credentials / Licences / Registrations
    ├── Engagement Type (employee / contract / gig / platform / freelancer ...)
```

├── Work Mode (onsite / field / hybrid / remote / home-based)
├── Seniority / Responsibility Level
├── Geography / Mobility / Service Radius
└── Market Tags (blue / white / grey / pink / hazardous-operations)

3. Collar labels: database treatment

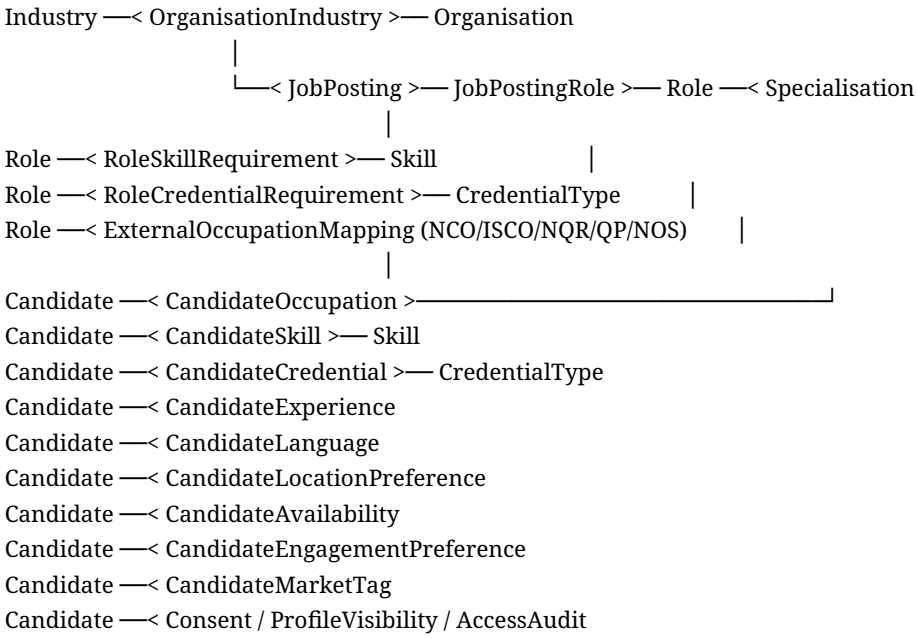
Label	Practical meaning	Example roles	Canonical?	Recommended field	Reason
Blue	Physical/ manual/ operational or trade work	Driver, delivery, housekeeping, security, cook	No	workforce_segment_tag	Broad descriptive class; roles span multiple skill levels [S13].
White	Office, professional, analytical, administrative or managerial work	Accountant, developer, HR, finance, engineer	No	workforce_segment_tag	Descriptive class, not occupation code [S14].
Pink	Historically female-dominated care/service/front-office work	Nursing, teaching, reception, salon, hospitality	No	legacy_market_tag	Gender-associated terminology; avoid as hiring hierarchy [S15].
Grey	Hybrid hands-on + technical/cognitive work	Lab technician, field technician, hardware support, paramedic	No	workforce_segment_tag	Useful commercial tag but boundaries overlap blue/white.
Black	Ambiguous label sometimes used for dirty/hazardous work, mining/oil, or other meanings	Mining/oil/ heavy hazardous operations	No	do_not_use_as_canonical	No stable authoritative Indian definition; model hazard/industry explicitly.
Gig	Work arrangement outside traditional employer-employee relationship	Driver, delivery, designer, developer, beautician	No	engagement_type	Statutory/ social-security concept; occupation-independent [S8].
Platform	Subset of gig	Ride-hailing,	No	engagement_ch	Platform is

Label	Practical meaning	Example roles	Canonical?	Recommended field	Reason
	work mediated by online platform	food delivery, task platform		annel	channel/arrangement, not role [S7][S8].

4. Independent axes required for GraphQL

- **Occupation:** What work is performed? Store as Role + specialization + NCO mapping.
- **Industry:** What business activity does the employer perform? Store as NIC-2025 mapping.
- **Competency:** What capability level is required? Store as NSQF/QP/NOS and role-skill requirements.
- **Engagement:** What contractual/work relationship applies? Store as Permanent, fixed-term, contract, gig, platform, freelancer, apprenticeship/articleship.
- **Work mode:** Where/how is work performed? Store as Onsite, field, remote, hybrid, home-based.
- **Work context:** In what service/use context? Store as Corporate chauffeur, ride-hailing, last-mile delivery, ICU, pharma sales.
- **Credential:** What permission/registration is required? Store as Driving licence, nursing registration, professional registration, trade certificate.
- **Geography:** Where can the candidate work? Store as Current city, PIN, radius, preferred cities, relocation, service radius.
- **Availability:** When can the person work? Store as Notice period, shifts, weekdays/weekends, immediate availability.
- **Compensation:** How is work paid? Store as Monthly, hourly, daily, shift, trip, task, delivery, commission.
- **Market segment:** How does the marketplace describe the work? Store as Blue/white/grey/service/pink legacy tag; never root hierarchy.

5. Relational entity model



5.1 Core table keys

JobFamily — id, code, name, parent_id?, active, taxonomy_version
Commercial navigation family; can evolve without changing NCO mapping.

Role — id, family_id, canonical_name, slug, description, active, valid_from/to
Canonical role; do not encode city/work mode/gig into role identity.

RoleAlias — role_id, alias, language_code, region_code, alias_type, weight
Supports "delivery boy", "rider", "bai", "peon" while keeping neutral canonical titles.

Specialisation — id, role_id, name, code, parent_id?, active
LMV Driver, ICU Nurse, GST Accountant, AC Technician.

ExternalOccupationMapping — role_id/specialisation_id, scheme, external_code, title, version, nsqf_level?, source_url, verified_at
Versioned mapping to NCO/ISCO/NQR/QP/NOS.

Skill — id, canonical_name, skill_type, parent_id?, active
Technical/tool/equipment/process/safety/service/digital/language/domain.

RoleSkillRequirement — role_id, specialisation_id?, skill_id, importance, mandatory, min_proficiency
Reusable structured matching criteria.

CandidateOccupation — candidate_id, role_id, specialisation_id?, is_primary, priority, years_experience, current_flag, verified
Many-to-many; supports e-Shram-like primary/secondary occupations [S9].

CandidateSkill — candidate_id, skill_id, proficiency, years_experience, last_used, verified
Never use comma-separated skills.

CandidateCredential — candidate_id, credential_type_id, issuer, number_tokenized/encrypted, issue/expiry, verification_status
Licences/registrations are independently verifiable and expirable.

EngagementType — code, name, legal_basis?, active
Permanent, fixed-term, contract, gig, platform-gig, freelancer, apprentice, articleship, self-employed.

WorkMode — code, name
ONSITE, FIELD, HYBRID, REMOTE, HOME_BASED.

CandidateLocationPreference — candidate_id, geography_id, radius_km?, willing_to_relocate, priority
Location-independent occupation model.

CandidateConsent — candidate_id, purpose, version, status, granted_at, withdrawn_at, expires_at
Candidate database access must be purpose/consent aware [S16].

CandidateAccessAudit — organisation_id, employer_user_id, candidate_id, action, timestamp
Audit SEARCH_RESULT / PROFILE_VIEW / CONTACT_REVEAL / EXPORT events.

6. Refined mapping of supplied job categories

Source label	Canonical family	Recommended subtypes / specialisations	Key structured skills / attributes	Market tag
Accountant	Accounting & Finance	Junior Accountant; AP; AR; GST/TDS; Payroll; Billing; Audit Support	Tally/ERP, Excel, reconciliation, GST/TDS, bookkeeping	White
Articleship	Professional Training	CA Articleship; Audit; Tax; Accounts; Industrial Training	Accounting, audit, tax, Excel; training status	White
Back Office	Business Operations & Administration	Operations; Documentation; KYC; Order Processing; MIS	Excel, documentation, CRM/ERP, email	White
Beautician	Beauty & Wellness	Beauty Therapist; Makeup; Hair; Nails; Spa	Skin/hair/makeup, hygiene, equipment	Service / Pink-tag
Business Development	Sales & Business Development	Inside BD; B2B BD; Partnerships; Channel; Lead Gen	Prospecting, CRM, negotiation, presentation	White
Computer Operator	Business Operations & Administration	Office/MIS/POS/ Billing/DTP/ERP Operator	Typing, Excel, Windows, scanning/printing, ERP/POS	White / Grey
Content Writing	Marketing & Content	SEO; Copywriting; Technical; Social; Script; Product Content	Writing, editing, SEO, CMS, research	White
Cook	Hospitality & Food Service	Domestic; Commis; Indian; South Indian; Chinese; Continental; Tandoor; Bakery	Food prep, hygiene, cuisine specialization	Blue / Service
Counsellor	Education & Training	Admission; Education; Career; Student; Healthcare; Customer	Communication, needs assessment, CRM	White / Service
Counter Sales	Sales & Business Development	Showroom; Pharmacy; Electronics; Jewellery; Apparel; QSR	POS, billing, product knowledge, upselling	Service / Pink-tag
Data Entry	Business Operations & Administration	Form Processing; Catalogue; KYC; Digitisation	Typing, accuracy, Excel, validation	White / Service
Delivery	Logistics &	Food; Grocery; Q-	Navigation,	Blue; Gig-capable

Source label	Canonical family	Recommended subtypes / specialisations	Key structured skills / attributes	Market tag
	Delivery	commerce; Parcel; Courier; Medicine; B2B	smartphone, POD, COD	
Digital Marketing	Marketing & Content	SEO; SEM/PPC; Performance; Social; Email/CRM; Affiliate	Ads platforms, analytics, SEO, optimization	White
Driver	Transport & Driving	LMV; 3W; LCV/Tempo; HGV; Bus; School Bus; Ambulance; Specialized	Licence, navigation, defensive driving, inspection	Blue / Grey; Gig-capable
Engineer	Engineering & Technical Services	Mechanical; Civil; Electrical; Electronics; Production; Quality; Site; Maintenance	CAD, drawings, QA/QC, domain tools	White / Grey
Finance	Accounting & Finance	Credit; Loan Processing; Collections; Insurance Ops; Analyst	Excel, credit, KYC, collections, financial analysis	White
Field Sales	Sales & Business Development	FMCG; Pharma; BFSI; Telecom; Real Estate; B2B	Territory, order booking, collection, CRM	Service / White
Graphic Designer	Marketing & Content	Brand; Social; DTP; Packaging; Motion; Visual	Photoshop, Illustrator, Figma, Canva	White
Government	NOT AN OCCUPATION	Employer-sector attribute: Government/PSU; occupation remains separate	N/A	N/A
Hardware Engineer	Technology & Software	Desktop; Laptop; Network; Printer; CCTV; Server; Electronics Repair	Diagnostics, networking, OS, repair	Grey
Hotel Management	Hospitality & Food Service	Front Office; Reservations; F&B Ops; Guest Relations; Banquet; Housekeeping Ops	PMS, guest service, SOPs	White / Service
Hotel / Waiter	Hospitality & Food Service	Steward; Waiter; Captain; Barista; Banquet; Room	POS, service, order taking, guest handling	Service / Pink-tag

Source label	Canonical family	Recommended subtypes / specialisations	Key structured skills / attributes	Market tag
		Service		
Housekeeping	Facilities & Security	Hotel; Office; Hospital; Industrial; Facility Cleaner; Supervisor	Cleaning SOPs, chemicals, equipment, hygiene	Blue / Service
HR Executive	Business Operations & Administration	Recruiter; TA; HR Ops; Payroll; Generalist; L&D; ER	ATS, sourcing, interviewing, HRMS	White
Lab Technician	Healthcare - Nursing & Allied	Medical Lab; Pathology; Sample Collection; Phlebotomy; Microbiology	Specimen handling, lab equipment, biosafety	Grey
Maid	Domestic & Home Services	Domestic Helper; Live-in; Part-time; Cook-cum-helper; Laundry	Cleaning, laundry, kitchen support	Blue / Service
Marketing	Marketing & Content	Brand; Trade; Field Promotion; Events; Product; Research	Campaigns, research, analytics	White
Medical	Healthcare - Medical	General Physician; Specialist Doctor; Dentist (separate regulated occupations)	Degree, registration, clinical specialty	White
Nursing	Healthcare - Nursing & Allied	Staff Nurse; ANM; GNM; BSc; ICU; OT; ER; Dialysis; Maternity; Home-care	Patient care, clinical documentation, registration	Grey / Service
Office Boy	Business Operations & Administration	Office Support; Pantry; Runner; Messenger; Dispatch	Office support, filing, dispatch	Blue / Service
Peon	Business Operations & Administration	Canonicalize as Office Support Attendant; retain Peon as alias	Document movement, errands, filing	Blue / Service
Receptionist	Business Operations & Administration	Corporate; Clinic; Hospital; Hotel; Salon; School	Calls, visitors, scheduling, basic computer	Service / Pink-tag
Retail	Retail & Store Operations	Sales Associate; Cashier; Store Associate;	POS, inventory, merchandising, service	Service / Pink-tag

Source label	Canonical family	Recommended subtypes / specialisations	Key structured skills / attributes	Market tag
		Merchandiser; Inventory; Picker/Packer; Supervisor		
Sales Executive	Sales & Business Development	Inside; B2C; B2B; Telesales; Showroom; Account Executive	Pitching, CRM, negotiation, closing	White / Service
Security Guard	Facilities & Security	Unarmed; Armed; Corporate; Residential; Retail; Industrial; Supervisor; CCTV	Access control, patrol, incident response	Blue
Software Developer	Technology & Software	Frontend; Backend; Full-stack; Android; iOS; .NET; Java; Python; Node; DevOps	Languages/ frameworks, Git, APIs, DBs	White
Teacher	Education & Training	Pre-primary; Primary; Secondary; Subject; Special Educator; Tutor; Online	Pedagogy, subject knowledge, classroom/digital teaching	White / Pink-tag
Technician	Engineering & Technical Services	Electrician; Plumber; HVAC; Appliance; Auto; Machine; Solar; Telecom; Field Service	Diagnostics, tools, safety, trade skills	Grey / Blue
Telecalling/BPO	Business Operations & Administration	Inbound; Outbound; Collections; Lead Gen; Domestic/International Voice; Chat/Email; Tech Support	Communication, CRM, ticketing, typing	Service / Pink-tag
Trainer	Education & Training	Corporate; Process; Sales; Soft Skills; Technical; Vocational	Facilitation, assessment, instructional design	White / Service
Other	UNMAPPED INTAKE	Temporary value only; taxonomy review must map it to a canonical	Raw title + mapping workflow	N/A

Source label	Canonical family	Recommended subtypes / specialisations	Key structured skills / attributes	Market tag
		role		

7. Worked example: Driver taxonomy

JobFamily: Transport & Driving
Role: Driver
- Specialisation: Car / LMV Driver
  - WorkContext: Personal Driver
  - WorkContext: Corporate Chauffeur
  - WorkContext: Taxi / Cab
  - WorkContext: Ride-hailing
- Specialisation: Three-Wheeler Driver
  - Auto-rickshaw
  - E-rickshaw
  - Goods three-wheeler
- Specialisation: Light Commercial Vehicle / Tempo
  - Pickup / Mini-truck
  - Last-mile distribution
- Specialisation: Heavy Goods Vehicle
  - Truck / Trailer / Container / Tanker
  - Long-haul
- Specialisation: Passenger Vehicle
  - Bus / Staff Bus / School Bus
  - Tourist Coach

Parallel attributes: licence class, vehicle/equipment, own vehicle, shift, outstation, route familiarity, languages, engagement type, platform experience, city/radius.

8. GraphQL schema strategy

GraphQL should expose the independent axes rather than mirror SQL tables one-for-one. Use stable IDs/codes, input objects for filters, cursor pagination, and explicit qualification/credential filters. Keep taxonomy lookups cacheable and versioned.

```
enum EngagementType { PERMANENT FIXED_TERM CONTRACT GIG PLATFORM_GIG FREELANCE APPRENTICESHIP ARTICLESHIP SELF_EMPLOYED }
enum WorkMode { ONSITE FIELD HYBRID REMOTE HOME_BASED }
enum MarketSegmentTag { BLUE WHITE GREY SERVICE PINK_LEGACY HAZARDOUS_OPERATIONS }

input CandidateSearchInput {
  roleIds: [ID!]
  specialisationIds: [ID!]
  skillFilters: [SkillRequirementInput!]
  credentialFilters: [CredentialRequirementInput!]
  engagementTypes: [EngagementType!]
  workModes: [WorkMode!]
  marketTags: [MarketSegmentTag!]
  location: LocationRadiusInput
```

```
  languages: [LanguageRequirementInput!]
  minExperienceMonths: Int
  maxNoticePeriodDays: Int
  shiftIds: [ID!]
  compensation: CompensationFilterInput
  verifiedOnly: Boolean
  taxonomyVersion: String
}

type Query {
  jobFamilies(version: String): [JobFamily!]!
  roles(familyId: ID, query: String, first: Int = 50, after: String): RoleConnection!
  candidateSearch(input: CandidateSearchInput!, first: Int = 25, after: String): CandidateConnection!
}

type CandidateConnection { edges: [CandidateEdge!]! pageInfo: PageInfo! totalCount: Int }
```

8.1 Query execution order

1. 1. Apply hard eligibility: role/specialisation, active credential, geography/radius, work mode, shift, availability.
2. 2. Apply structured skill constraints and experience thresholds.
3. 3. Apply optional semantic similarity over title/experience text only after structured eligibility.
4. 4. Rank by verified evidence, role relevance, required-skill coverage, experience, availability freshness and distance.
5. 5. Enforce organisation subscription entitlements and candidate privacy before returning PII/contact details.

8.2 Performance rules for GraphQL

- Use cursor/keyset pagination, not deep OFFSET pagination, for candidateSearch.
- Batch role/skill/credential relations with DataLoader-style resolvers to avoid N+1 queries.
- Denormalize only search projections: maintain a CandidateSearchDocument / search index fed from canonical relational data.
- Use immutable taxonomy IDs; update names/aliases through versioned records rather than recycling IDs.
- Index CandidateOccupation(role_id, specialisation_id, candidate_id), CandidateSkill(skill_id, candidate_id), active credentials, geo cells/PIN and availability.
- Separate public profile fields from protected contact PII; contact reveal should be a mutation/event with entitlement + audit.
- Persist saved searches as normalized filter JSON plus taxonomy version to allow deterministic reruns.

9. Taxonomy governance and versioning

- TaxonomyVersion(id, semantic_version, effective_from, status, release_notes).
- Never delete used roles; mark inactive/deprecated and map successor_role_id.
- External mappings must include framework version because NCO/NIC/QP/NOS revisions can change independently.

- “Other” is an intake state only. Every unmatched title enters a review queue and receives a canonical mapping or a new governed role.
- RoleAlias is region/language-aware. Search aliases can include colloquial Indian terms while canonical labels remain neutral/professional.
- Use one Role across cities; geography is always a separate filter. Metro/Tier-1/Tier-2/Tier-3 is a mutable market attribute, not role identity.

10. Foundation rules to lock before implementation

RULE-01 Occupation is the core; collar is a tag.

RULE-02 Industry (NIC) and occupation (NCO) are independent.

RULE-03 Gig/platform is engagement/channel, never a parent occupation.

RULE-04 Candidate ↔️ occupation and candidate ↔️ skill are many-to-many.

RULE-05 Work From Home is WorkMode=REMOTE/HOME_BASED, never a duplicate role.

RULE-06 Government is EmployerSector, never an occupation.

RULE-07 Articleship/apprenticeship are training/engagement dimensions linked to a functional occupation.

RULE-08 Credentials are structured, expirable and verifiable.

RULE-09 Aliases/local-language terms are separate from canonical role names.

RULE-10 Search is structured-first, semantic-second.

RULE-11 Taxonomy and external mappings are versioned.

RULE-12 Candidate PII access is consent-, entitlement- and audit-controlled.

11. Research references

Sources were re-checked for this 25 September 2026 architecture pass. Government/ILO sources are treated as authoritative for formal classifications; dictionary/market sources are used only to interpret informal collar terminology.

[S1] Ministry of Labour & Employment, National Classification of Occupations 2015, Volume I. NCO-2015 structure: 8-digit hierarchy, alignment to ISCO-08, linkage to QP/NOS/NSQF. https://labour.gov.in/sites/default/files/National%20Classification%20of%20Occupations%20_Vol%20I-%202015.pdf

[S2] Ministry of Labour & Employment, National Classification of Occupations 2015, Volume II-B. NCO covers 3,600 civilian occupations across 52 sectors; describes duties and skill requirements.

https://labour.gov.in/sites/default/files/National%20Classification%20of%20Occupations_Vol%20II-B-%202015.pdf

[S3] MoSPI, National Industrial Classification 2025. Current Indian industry/economic activity taxonomy; includes concordance with NIC-2008. https://www.mospi.gov.in/sites/default/files/main_menu/national_industrial_classification/NIC_2025Final_18112025.pdf

[S4] National Qualifications Register, NSQF Gazette Notification 2023. NSQF levels 1-8 with intermediate levels; competency/outcome descriptors. https://www.nqr.gov.in/downloads/pdfs/NSQF_Gazette_Notification.pdf

[S5] NSDC, Sector Skill Councils. Sector Skill Councils identify skill needs and develop competency standards/qualifications. https://nsdcindia.org/sector-skill-councils

[S6] NSDC, National Occupational Standards and Model Curriculum. NOS define workplace functions; sets of NOS aligned to a job role form a Qualification Pack. https://www.nsdcindia.org/national-occupational-standards-and-model-curriculum

[S7] NITI Aayog, India's Booming Gig and Platform Economy. Gig workforce estimates/projections and analysis of platform/non-platform work. https://www.niti.gov.in/node/375

[S8] Ministry of Labour & Employment, Code on Social Security 2020. Statutory definitions of gig worker, platform worker, home-based worker and fixed-term employment. https://labour.gov.in/sites/default/files/ss_code_as_passed_by_lok_sabha.pdf

[S9] e-Shram FAQ. Distinguishes occupation from skill and recognizes primary and secondary occupations. https://eshram.gov.in/faqs

[S10] e-Shram Portal. National database of unorganised workers includes occupation, occupation type, education and skill types, including gig/platform workers. https://eshram.gov.in/e-shram-portal

[S11] ILO, ISCO-08. International occupation hierarchy and definitions of job, occupation, skill level and skill specialization. https://isco.ilo.org/en/isco-08/

[S12] ILO, Occupational Classification Tools & Resources. Occupation classification should use tasks/duties as core variable; similarity criteria include skill level and specialization. https://isco.ilo.org/en/resource/

[S13] Cambridge Dictionary, blue-collar. Common-language definition of physical/manual work; demonstrates collar labels are descriptive rather than formal Indian occupational codes. https://dictionary.cambridge.org/dictionary/english/blue-collar

[S14] Cambridge Dictionary, white-collar. Common-language definition of office/mental work. https://dictionary.cambridge.org/dictionary/english/white-collar

[S15] Cambridge Dictionary, pink-collar. Historical gender-associated label; should not be a canonical hierarchy axis. https://dictionary.cambridge.org/us/dictionary/english/pink-collar

[S16] MeitY, Digital Personal Data Protection Rules 2025. Current privacy-rule publication page relevant to candidate data processing and access controls.
https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa?pageTitle=Digital-Personal-Data-Protection-Rules-2025%C3%AF%C2%BF%C2%BC

Appendix A. SQL Server starter DDL (conceptual)

```
CREATE TABLE dbo.Role (
   Id bigint IDENTITY PRIMARY KEY,
   JobFamilyId bigint NOT NULL,
   CanonicalName nvarchar(200) NOT NULL,
   Slug nvarchar(200) NOT NULL,
   TaxonomyVersionId bigint NOT NULL,
   IsActive bit NOT NULL DEFAULT 1,
   ValidFrom datetime2 NOT NULL DEFAULT sysutcdatetime(),
   ValidTo datetime2 NULL
);

CREATE TABLE dbo.CandidateOccupation (
   CandidateId bigint NOT NULL,
   RoleId bigint NOT NULL,
   SpecialisationId bigint NULL,
   IsPrimary bit NOT NULL DEFAULT 0,
   Priority tinyint NULL,
   ExperienceMonths int NULL,
   IsCurrent bit NOT NULL DEFAULT 0,
   VerificationStatus tinyint NOT NULL DEFAULT 0,
   CONSTRAINT PK_CandidateOccupation PRIMARY KEY (CandidateId, RoleId, SpecialisationId)
);

CREATE INDEX IX_CandidateOccupation_Search
ON dbo.CandidateOccupation(RoleId, SpecialisationId, ExperienceMonths, CandidateId);
```